CPP Lab 2 Assignment

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Roll NO:20 BSCS Section A

LAB Task-1

A:- Input cpp

```
# include <iostream>
using namespace std;
int main () {
char Letter;
Letter = 'A';
cout << letter << '\n';
letter = 'B';
Cout letter '\n';
Cin >> letter;
return 0;
}
```

Output

A

B

B:- Input CPP

```
# include < iostream>
using name space std;
int main () {
char letter;
letter = 65;
Cout << letter << '\n';
letter = 66;
Cout << letter << '\n';
Cin >> letter;
return 0;
}
```

Output

65

66

Task 2:-

-) **Source code (cpp)**

```
# include <iostream>
using name space std;
int main (){
Char letter;
letter = 'A';
Cout<< letter<< '\n';
letter = 'B';
Cout << letter<<'\n';
Cin >> letter;
return 0;
}
```

Task 3:-

Source code (cpp)

```
# include <iostream>
using name space std;
int main (){
int val1;
int val2;
int val3;
Cout <<"Please enter your 3 numbers";
Cin >> val1 >> val2 >> val3;
Cout << val1<< "\n" << val2<< "\n" <<val3 << "\n";
Cout << val3<< "\n" << val2<< "\n" << val1 << "\n";
return 0;
}
```

Task 4:-

Source Code (.cpp)

```
# include <iostream>
# include <cmath>
using name space std;
int main() {
double x1, y1, x2, y2;
Cin >> x1 >> y1 >> x2 >> y2;
double d = sqrt (pow (x2 - x1, 2) + (pow(y2-y1);
cout << d;
return 0;
}
```

Task-5 :-

Input cpp:

```
#include <iostream>
using namespace std;
int main (){
double r, l, w, b, h;
double PI = 3.14159;
Cout << " Enter the radius of circle";
Cin >> r;
Cout << " Area of circle =" << PI * r * r << endl;
Cout << " Enter Length and wight of rectangle :";
Cin >> l >> w;
Cout << " Area of rectangle = " << l * w << endl;
Cout << " Enter the base and height of triangle:";
Cin >> b >> h;
Cout << " Enter Area of Triangle=" << b * h * 0.5 << endl;
return 0;
}
```

Task-6 :-

Source Code (cpp)

```
* include <iostream>
using namespace std;
int main () {{
double p1 = 12.95, p2 = 24.95, p3 = 6.95, p4 = 14.95, p5 = 3.95;
double subtotal = p1 + p2 + p3 + p4 + p5;
double tax = subtotal + tax;
cout << "price of item 1 = $" << p1 << endl;
cout << " price of item 2 = $" << p2 << endl;
cout << " price of item 3 = $" << p3 << endl;
cout << " price of item 4 = $" << p4 << endl;
cout << " price of item 5 = $" << p5 << endl;
Cout << "In subtotal = $" << subtotal << endl;
cout << "sales tax (6%) = $" << tax << endl;
cout << "Total = $" << total << endl;
return 0;
}
```

Output :-

price of item 1 = $ 12.95
price of item 2 = $ 24.95
price of item 3 = $ 6.95
price of item 4 = $ 14.95
price of item 5 = $ 3.95

Subtotal = $ 63.75

Sales tax (6%) = $ 3.825

= $ 67.575

Bonus Task :-

Source Code (cpp)

```
# include < iostream>
using namespace std;
int main() {
Cout<<" Enter a five digit Integer:";
Cin >> num;
int d1 = num /10000;
int d2 = (num /1000) % 10;
int d3 = (num /1000) % 10;
int d4 = (num /10) % 10;
int d5 = (num%10);
Cout << d1 <<" " <<d2 <<" " << d3<<" "<<d4<<" "<<d5<<endl
return 0;
}
```

Home Task :-

Task 1 :-

Source Code (cpp)

```
# include < iostream>
using name space std;
int main (){
int var1, var2;
Cin >> var1 >> var 2;
Swap (var1, var2);
Cout << "var=" << var1 << " var 2=" var 2;
}
```

Task 2 :-

Source Code (cpp)

```
# include < iostream >
using namespace std;
int main (){
Char ch;
Cin >> ch;
Cout << char ( touppet(ch));
}
```

```cpp
#include <iostream>
using namespace std;

int main() {
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

```
A
B
65

[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int val1;
    int val2;
    int val3;

    cout << "Please enter your 3 numbers";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    return 0;
}
```


Please enter your 3 numbers

```cpp
#include <iostream>
#include <cmath>
using namespace std;

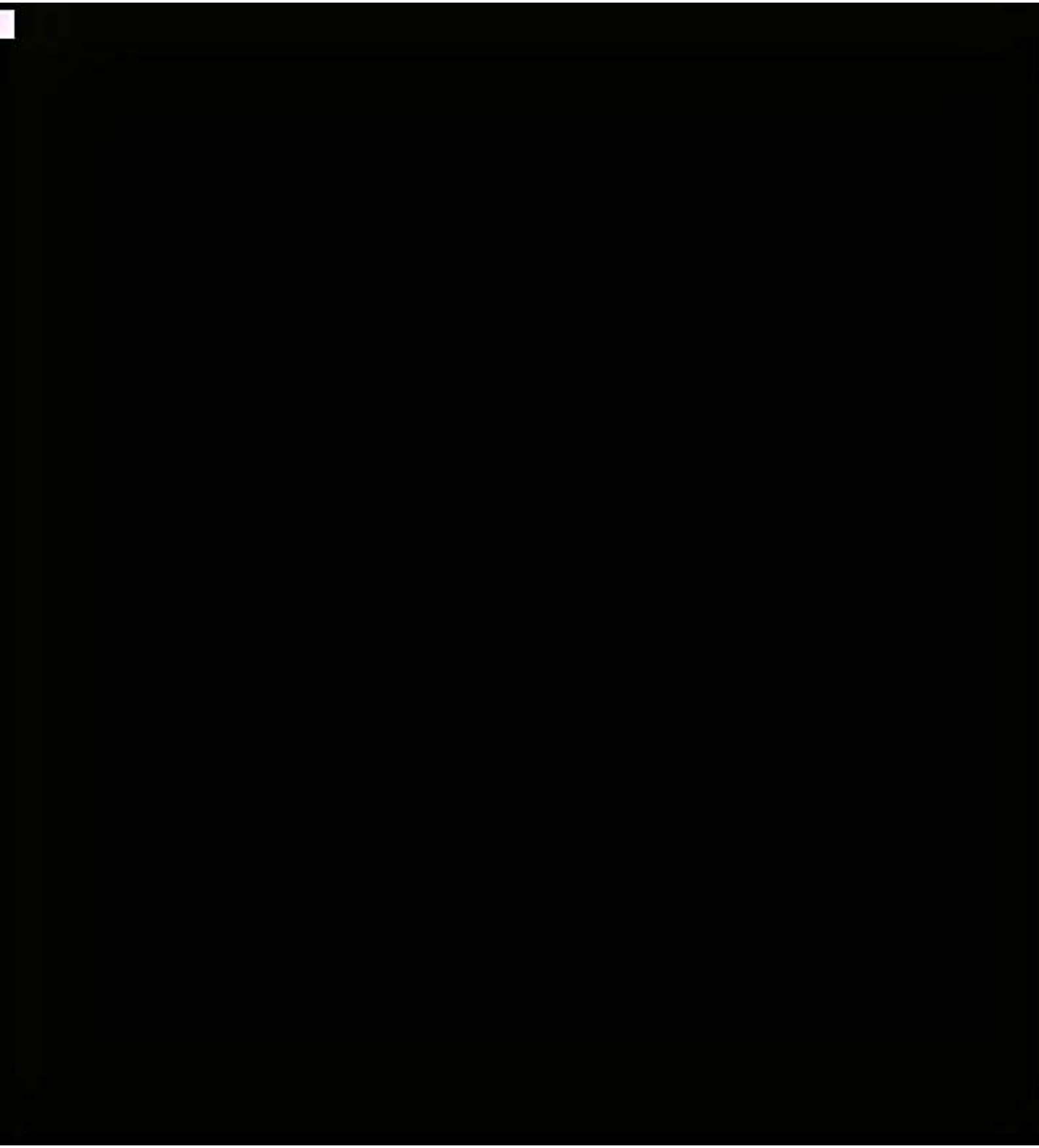
int main() {
    double x1, y1, x2, y2;

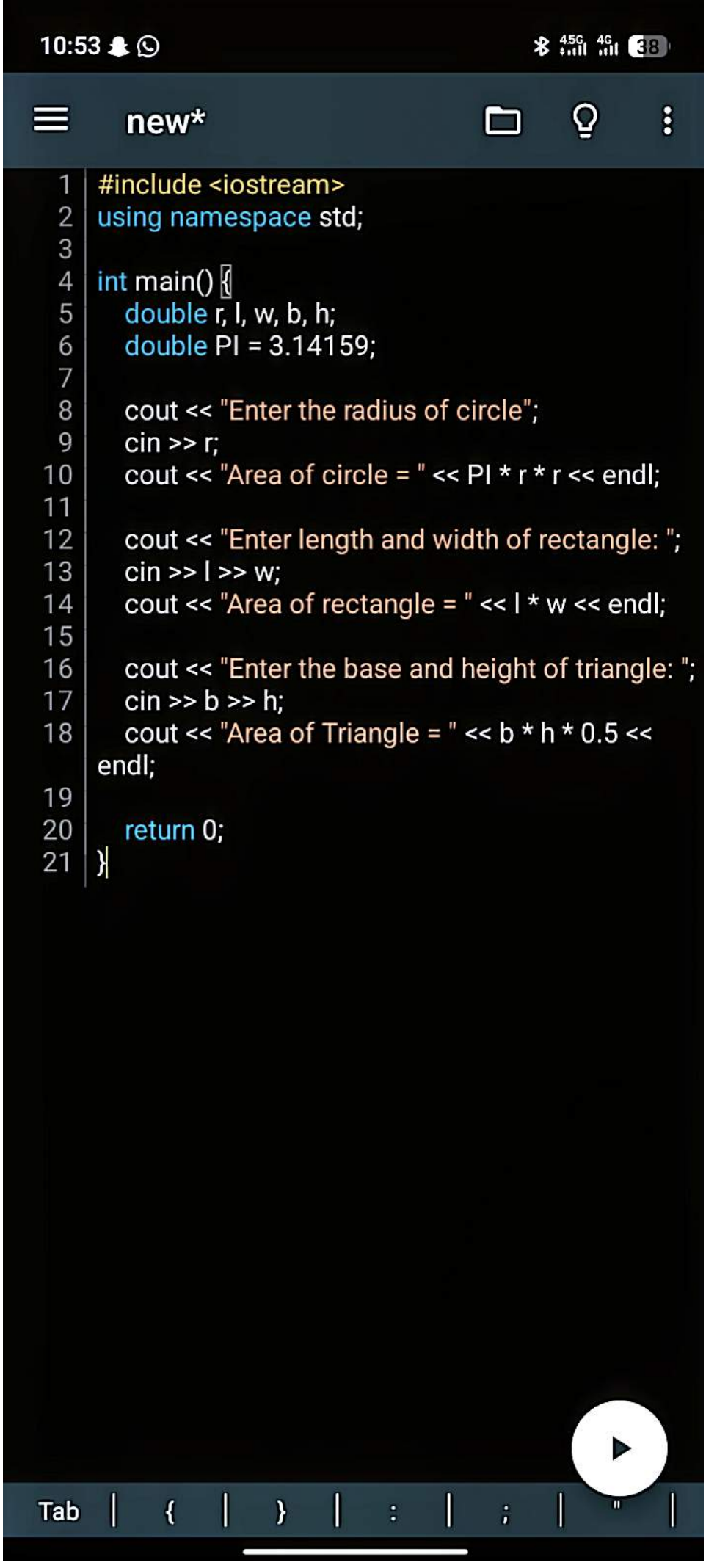
    cin >> x1 >> y1 >> x2 >> y2;

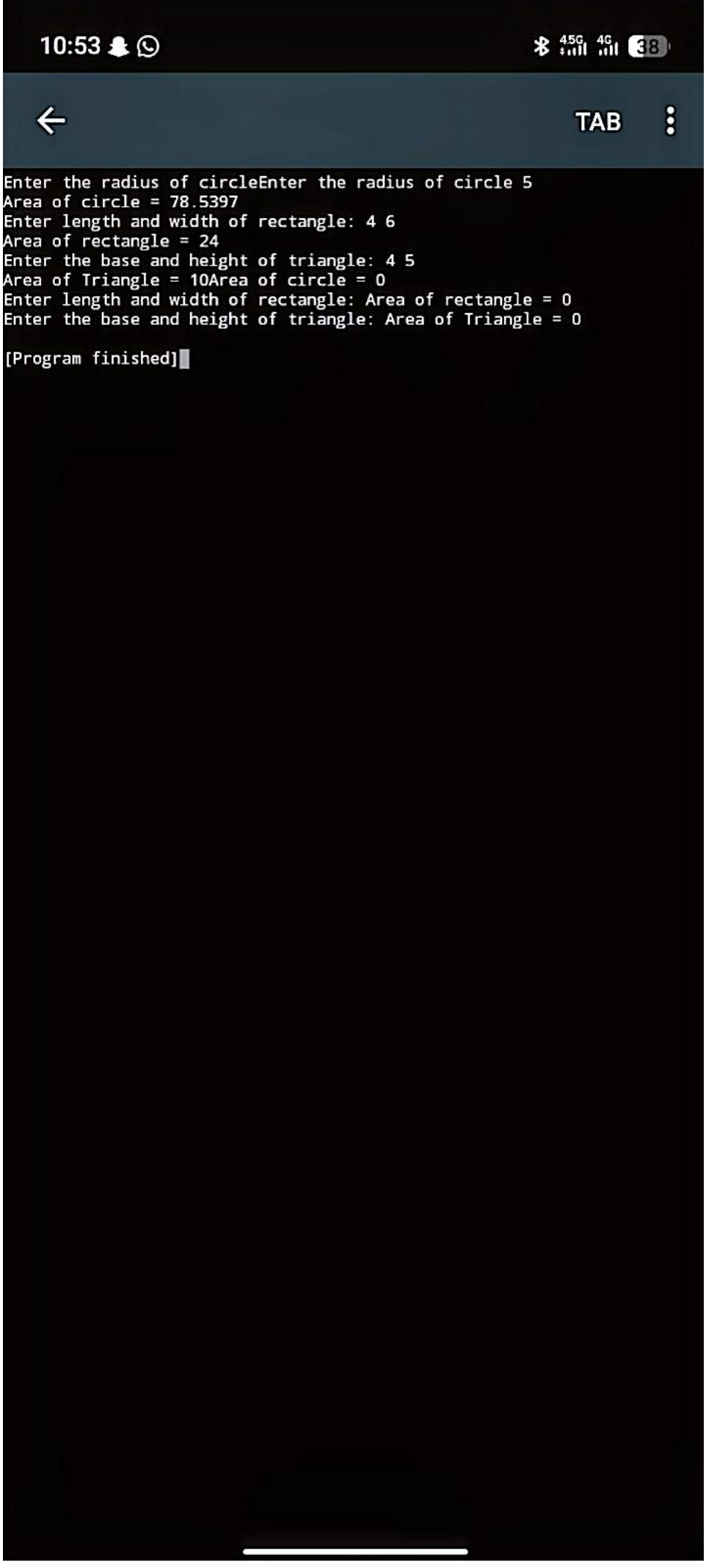
    double d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

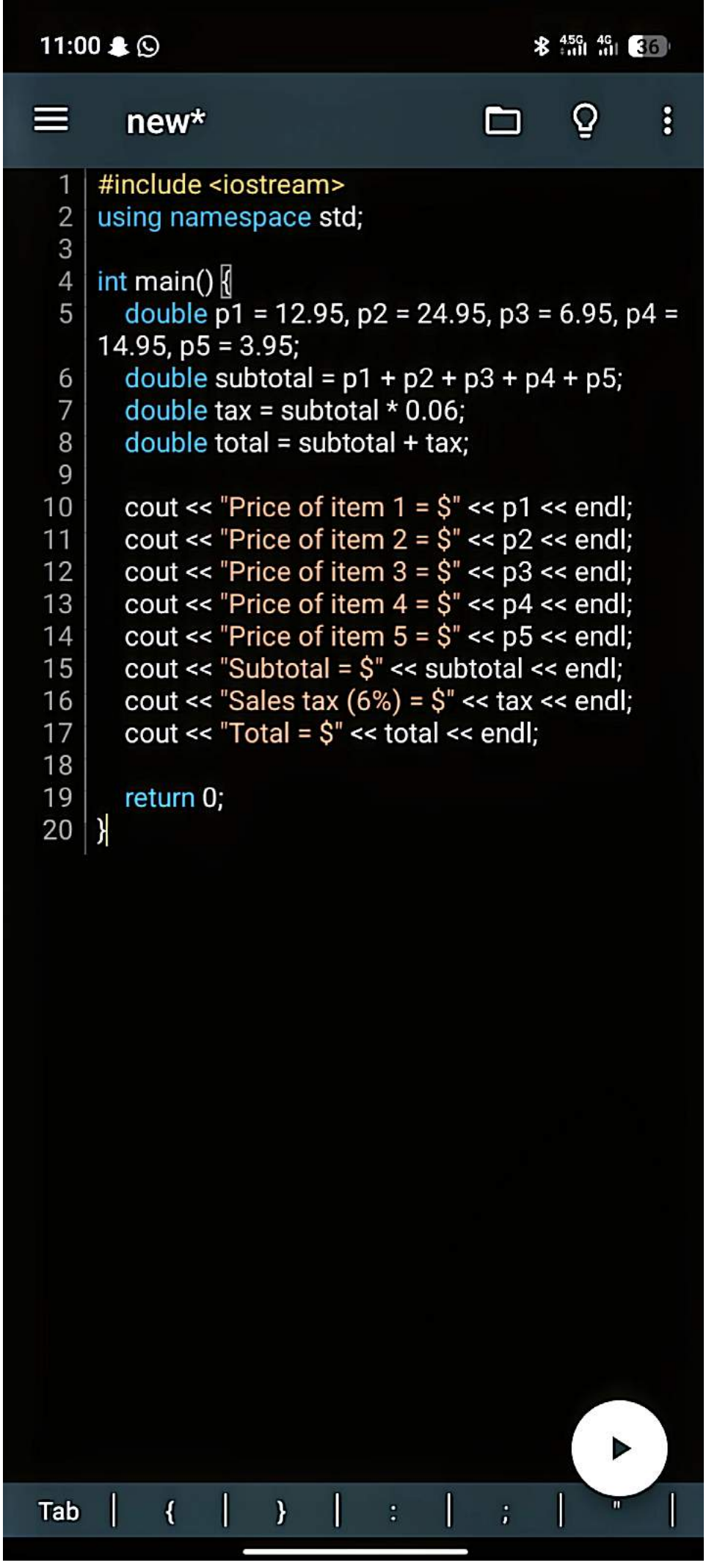
    cout << d;

    return 0;
}
```




10:53
new*
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5 double r, l, w, b, h;
6 double PI = 3.14159;
7
8 cout << "Enter the radius of circle";
9 cin >> r;
10 cout << "Area of circle = " << PI * r * r << endl;
11
12 cout << "Enter length and width of rectangle: ";
13 cin >> l >> w;
14 cout << "Area of rectangle = " << l * w << endl;
15
16 cout << "Enter the base and height of triangle: ";
17 cin >> b >> h;
18 cout << "Area of Triangle = " << b * h * 0.5 <<
endl;
19
20 return 0;
21 }
Tab
{
}
:
;
"


10:53
TAB
Enter the radius of circleEnter the radius of circle 5
Area of circle = 78.5397
Enter length and width of rectangle: 4 6
Area of rectangle = 24
Enter the base and height of triangle: 4 5
Area of Triangle = 10Area of circle = 0
Enter length and width of rectangle: Area of rectangle = 0
Enter the base and height of triangle: Area of Triangle = 0
[Program finished]


11:00
new*
#include <iostream>
using namespace std;
int main() {
double p1 = 12.95, p2 = 24.95, p3 = 6.95, p4 = 14.95, p5 = 3.95;
double subtotal = p1 + p2 + p3 + p4 + p5;
double tax = subtotal * 0.06;
double total = subtotal + tax;
cout << "Price of item 1 = $" << p1 << endl;
cout << "Price of item 2 = $" << p2 << endl;
cout << "Price of item 3 = $" << p3 << endl;
cout << "Price of item 4 = $" << p4 << endl;
cout << "Price of item 5 = $" << p5 << endl;
cout << "Subtotal = $" << subtotal << endl;
cout << "Sales tax (6%) = $" << tax << endl;
cout << "Total = $" << total << endl;
return 0;
}
Tab
{
}
:
;
"

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales tax (6%) = $3.825
Total = $67.575

[Program finished]
```